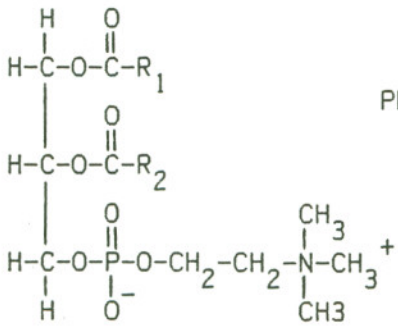
BIOCHEMISTRY CUMULATIVE EXAMINATION

January 19, 1991



Phosphotidyl Choline

Phospholipases are designated as phospholipase A if they catalyze the hydrolysis of either of the ester linkages between the fatty acids and glycerol, phospholipase C if they catalyze hydrolysis of the phosphoester linkage between glycerol and the phosphate, and D if they catalyze the hydrolysis of the bond between the phosphate and choline.

You have phosphatidyl choline labeled with P^{32}. At time 0 it has a specific radioactivity of 0.01 uCi/mMole. (One curie = 3.7×10^{10} decompositions per second.) Ten uMoles of substrate in 100 ul of the appropriate buffer was incubated with 10 ng of pure enzyme for 20 min. The enzyme has a molecular weight of 25,000. The reaction was stopped with heat, then the incubation mixture was applied to a cation exchange column in a low salt buffer. The nonretained fraction which eluted from the column had 1,370 decompositions per minute 24 hours after time 0. The rest of the radioactivity eluted with high salt. (HINT: Phosphatidyl choline sticks to the cation exchange column.) The half life of P^{32} is 14 days. You might need the following equations:

$$T_{1/2} = 0.693/k$$

$$\log[N/N_o] = kT/2.303$$

N is the number of atoms present at time T while N_o is the number of atoms at 0 time.

(20pts) 1. What type of phospholipase was used?

(40pts) 2. How many uMoles of product were formed?

(40pts) 3. What is the specific activity of the enzyme?